Building a Digital FTE

Mode 2 Phase 2 (Build Workers), Chapter 2/3 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/digital-fte-crash-course

EN: The previous course's agent forgot everything when the terminal closed, with Python-hardwired tools. This is the first real step from 'agent' to 'AI Worker': Abilities->Skills, restart-forgotten-state->Postgres (system of record), MCP = the wire connecting the agent to the store.
Roman Urdu: Pichle course ka agent terminal band karte hi sab bhool jata, tools Python-hardwired the. Yeh 'agent se AI Worker' ka pehla real qadam: Abilities->Skills, restart-forgotten-state->Postgres (system of record), MCP = wire jo agent ko store tak leti.

1. Skills -- Progressive Disclosure

- A skill = folder + SKILL.md, an open standard. **3-stage loading:** Discovery (~100 tokens/skill, all loaded) -> Activation (full body, on match) -> Execution (referenced files on-demand).
- **Description IS the trigger** -- the whole game is in the description: what it produces, when to reach for it, keywords, and a do-NOT line.

2. Neon Postgres + pgvector as System of Record

- 4 data kinds: business records, reference library (docs+embeddings), state (conversations), trace (audit_log). Stored and query embeddings must use the SAME model.
- **3 vector operators:** <=> cosine (text default), <-> straight-line (image), <#> dot product (rare). HNSW is the right default index.
- Audit test: given a conversation + timestamp, can you reconstruct what happened WITHOUT re-running the model? If not, it's just logs. Action + record happen in ONE transaction.

3. MCP -- Wiring the Agent to the Store

- MCP is NOT a framework, a service, a security boundary, or a @function_tool replacement -- it's an open client/server protocol ('USB-C for AI tools').
- Neon's own MCP server is a DEVELOPMENT plane tool, never production -- run_sql can run ANY SQL, a door onto a live DB.
- Build a CUSTOM, narrow MCP server when you need process isolation, tighter scope, or reusability across consumers -- not by default.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Skill (SKILL.md)	Folder + description-triggered instructions an agent loads progressively.	Folder + description-triggered instructions jo agent progressively load karta.
System of Record (Postgres)	The durable store replacing restart-forgotten in-memory state.	Durable store jo restart-forgotten in-memory state ki jagah leta.
MCP (Model Context Protocol)	An open client/server protocol connecting an agent to tools/data -- not a security boundary.	Open client/server protocol jo agent ko tools/data se connect karta -- security boundary nahi.
Audit trail test	Can you reconstruct what happened from the record alone, without re-running the model?	Kya aap sirf record se reconstruct kar sakte ho, model dobara chalaye bina?
Custom MCP server	Built when you need process isolation, tighter scope, or reuse -- not the default choice.	Jab process isolation, tighter scope, ya reuse chahiye -- default choice nahi.

Quick Self-Check

Question	Answer
Why is the skill's 'description' field called the whole game?	It's what fires at the Discovery stage, before the full skill body ever loads -- it alone decides whether the skill gets used.
Why must the stored embedding and the query embedding use the same model?	Different models produce incompatible vector spaces -- comparing them would give meaningless similarity scores.
Why must action and audit-record happen in one transaction?	If they're separate, a crash between them leaves an action with no record -- breaking the ability to reconstruct what happened.
Why is Neon's own MCP server unsafe for production use?	Its run_sql tool can execute any SQL -- a direct door onto a live database, meant only for the development plane.

Where This Leads / Sources

Leads to: Give Your AI Agent a Nervous System (the remaining 5 invariants). Full worked example (Customer Support Worker, 9 decisions) in Part 4.

Source: docs/digital-fte-crash-course/ (this repo), content originally via Zia Tutor AI connector.